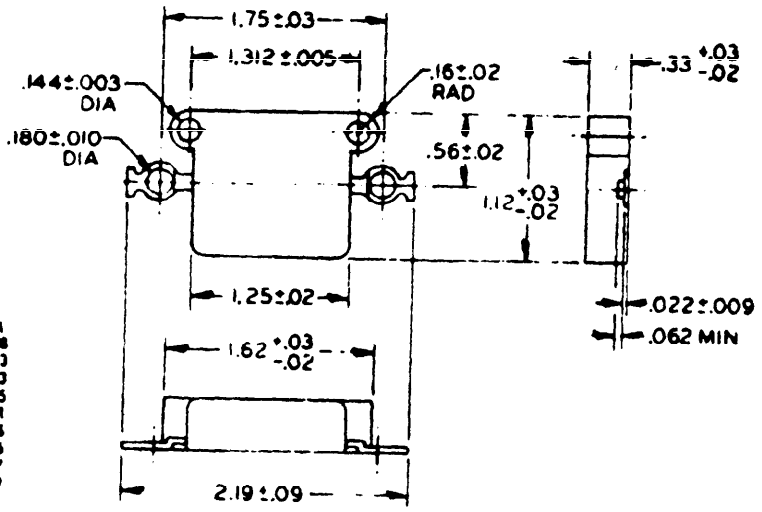
MIL-C-5/7F
12 AUGUST 1966
SUPERSEDING
MIL-C-5/7E
10 October 1961

MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, MICA DIELECTRIC

STYLE CM45

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-5.



INCHES	MM
.003	.08
.005	.13
.009	.23
.010	.25
.02	.51
.022	.56
.03	.76
.062	1.57
.09	2.29
.144	3.66
.16	4.06
.180	4.57
.3	8.38
.56	14.22
1.1	28.45
1.2	31.75
1.3 2	33.32
1.6.	41.15
1.75	44.45
2.19	55.63

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Approximate weight .62 ounce.

FSC 5910

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REQUIREMENTS:

Dimensions: See figure.

Capacitance value: See table I.

Capacitance tolerance: See table I.

DC rated voltage: See table I.

Operating temperature range: See table I.

Symbol	Temperature range
N	-55° C to +85° C
O	-55° C to +125° C
P	-55° C to +150° C

Vibration grade: 3 (10 to 2,000 cps).

Characteristic:

Symbol	Temperature coefficient PPM/° C	Capacitance drift
B	No specified	Not specified

Dielectric withstanding voltage:
Method 301 of MIL-STD-202:
200 percent of dc working voltage for not less than 1 second nor more than 5 seconds.

Barometric pressure:
Method 105 of MIL-STD-202:
Condition B (50,000 ft).
Test potential - 150 percent of rated dc voltage for not less than 1 second nor more than 5 seconds.

Insulation resistance:
Method 302 of MIL-STD-202, condition A.
At 25° C - 10,000 megohms minimum.
At 85° C - 4,000 megohms minimum.
At 125° C - 1,500 megohms minimum.
At 150° C - 750 megohms minimum.

Dissipation factor, maximum: See MIL-C-5.

Capacitance:
Method 305 of MIL-STD-202.

Vibration:
Method 204 of MIL-STD-202, condition D (20 G).

Temperature cycling:
Method 102 of MIL-STD-202, condition D.

Immersion:
Method 104 of MIL-STD-202, condition B.
Dielectric withstanding voltage - Meet initial requirement.
Insulation resistance - 30 percent of initial requirement.
Dissipation factor - not greater than 150 percent of initial requirement.
Capacitance change - Shall not exceed ±4 percent of nominal value of 1 pf, whichever is greater.

Shock (medium impact):
Method 204 of MIL-STD-202, condition C (50 G).

Moisture resistance:
Method 106 of MIL-STD-202:
Dielectric withstanding voltage - Meet initial requirement.
Insulation resistance - At least 30 percent of initial requirement.
Dissipation factor - not greater than 150 percent of initial requirement.
Capacitance change - Shall not exceed ±4 percent of nominal value or 1 pf, whichever is greater.

Temperature coefficient and capacitance drift:
Not specified.

Low temperature storage and life:
48 hours at -55 $^{+0}_{-3}$ °C, immediately followed by qualification or performance test, as applicable.
Qualification test - 2,000 hours at applicable high test temperature with 150 percent of dc working voltage applied.
Dielectric withstanding voltage - Initial requirements.
Insulation resistance - Initial requirements.
Dissipation factor - not greater than 150 percent of initial requirement.
Capacitance change - Shall not exceed ±5 percent of nominal value or 1 pf, whichever is greater.
Performance check - Conditions and requirements in accordance with qualification test except test duration shall be 250 hours.
Continuation test - Units that have been subjected to the 250 hour test shall be continued for an additional 1,750 hours and meet the requirements for the qualification test.

Qualification inspection:
Qualification inspection (Group B) is required.

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TABLE I. Style CM45.

Type designation 1/	Capacitance (pf)	DC working voltage (volts)	Operating temperature range	Characteristic available for capacitance tolerance	
				±2% (G)	±5% (J)
CM45BK470--3	47	2500	NOP	B	B
CM45BK510--3	51	2500	NOP	B	B
CM45BK560--3	56	2500	NOP	B	B
CM45BK620--3	62	2500	NOP	B	B
CM45BK680--3	68	2500	NOP	B	B
CM45BK750--3	75	2500	NOP	B	B
CM45BK820--3	82	2500	NOP	B	B
CM45BK910--3	91	2500	NOP	B	B
CM45BK101--3	100	2500	NOP	B	B
CM45BK111--3	110	2500	NOP	B	B
CM45BK121--3	120	2500	NOP	B	B
CM45BK131--3	130	2500	NOP	B	B
CM45BK151--3	150	2500	NOP	B	B
CM45BK161--3	160	2500	NOP	B	B
CM45BK181--3	180	2500	NOP	B	B
CM45BK201--3	200	2500	NOP	B	B
CM45BK221--3	220	2500	NOP	B	B
CM45BK241--3	240	2500	NOP	B	B
CM45BK271--3	270	2500	NOP	B	B
CM45BK301--3	300	2500	NOP	B	B
CM45BK331--3	330	2500	NOP	B	B
CM45BK361--3	360	2500	NOP	B	B
CM45BK391--3	390	2500	NOP	B	B
CM45BK431--3	430	2500	NOP	B	B
CM45BK471--3	470	2500	NOP	B	B
CM45BK511--3	510	2500	NOP	B	B
CM45BK561--3	560	2500	NOP	B	B
CM45BK621--3	620	2500	NOP	B	B
CM45BK681--3	680	2500	NOP	B	B
CM45BK751--3	750	2500	NOP	B	B
CM45BK821--3	820	2500	NOP	B	B
CM45BK911--3	910	2500	NOP	B	B
CM45BK102--3	1000	2500	NOP	B	B
CM45BK112--3	1100	2500	NOP	B	B
CM45BK122--3	1200	2500	NOP	B	B
CM45BK132--3	1300	2500	NOP	B	B
CM45BK152--3	1500	2500	NOP	B	B
CM45BK162--3	1600	2500	NOP	B	B
CM45BK182--3	1800	2500	NOP	B	B

1/ Complete type designation will include additional symbols to indicate capacitance tolerance and temperature range.

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Custodians:
Army - EL
Navy - SH
Air Force - 11

Review activities:
Army - EL
Navy - SH
Air Force - 11, 17, 85

User activities:
Army -
Navy -
Air Force - 14, 19

Preparing activity:
Army - EL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
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	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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